Chapter 27 — Secrets Management & Obfuscation (R8)

> **Part III — Authentication & Security** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 20 (Keystore — runtime secret protection), Ch 26 (integrity — what obfuscation complements), Ch 1 (networking — where API keys are used). Pairs with Ch 83 (crash reporting — mapping files).
>
> **Reference surface in StreamCart:** keeping API keys, third-party SDK secrets, and business logic out of attackers' reach in a decompilable APK — and knowing what's actually protectable on the client.

27.1 Problem Definition and Real-World Context

An APK is **shippable, decompilable code**. Anything baked into it — strings, classes, logic — can be extracted with standard tools (apktool, jadx) in minutes. This creates two linked problems: **where do secrets go**, and **how much does obfuscation actually protect**? Naive assumptions cause real breaches:

- **Hardcoded secrets.** API keys, signing secrets, or third-party credentials in code/resources/`BuildConfig` are trivially extracted from the APK — the single most common Android secret leak. "Obfuscating" a hardcoded secret doesn't help; a determined attacker finds it at runtime.
- **Treating obfuscation as encryption.** R8/ProGuard renames and shrinks code, raising the *cost* of reverse engineering — it is **not** encryption and does **not** hide secrets or logic from a motivated attacker. Over-trusting it is dangerous.
- **Client-side "secret" that must be secret.** Any truly-secret value (a key that grants backend access) **cannot** live safely on the client at all — it belongs **server-side**; the client should hold only revocable, scoped, low-value tokens (Ch 19) and fetch what it needs over authenticated channels.
- **Breaking crash reports / forgetting mapping files.** Enabling R8 without retaining the **mapping file** makes stack traces unreadable (Ch 83).

The architecture: **keep true secrets server-side**; for unavoidable client config, use **build-time injection from secure sources** (not VCS), **revocable/restricted keys**, **runtime protection via Keystore** (Ch 20) for secrets that must persist, **R8** for shrinking + raising reverse-engineering cost (not as security), and **defense-in-depth** (Ch 20/23/26). The core lesson: **you cannot keep a secret in an APK — so design so the client doesn't need to.**

Three practical use cases

1. **Third-party API keys (Maps, analytics).** Use **restricted** keys (package + signature + API restrictions) so extraction yields a key that only works for your app's scope and can be revoked/rotated.
2. **Backend access (StreamCart API).** No static API key in the app; the client authenticates per-user (Ch 19) and the backend authorizes — there is no "app secret" to steal.

3. **Business logic / anti-abuse (Wallet/Watch).** Obfuscate + move critical checks server-side; use integrity (Ch 26) rather than hiding logic in the client.

Why this is architectural

Secrets management defines **what can and cannot live on the client**, the **build-time secret pipeline**, the **runtime protection** (Keystore), and the honest **role of obfuscation** — properties that determine whether shipping the app leaks your keys and whether reverse engineering is cheap.

27.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	No true secrets hardcoded in code/resources/ `BuildConfig` /VCS.
F2	Inject necessary client config at build time from secure sources (not committed).
F3	Use restricted, revocable keys for unavoidable third-party client keys.
F4	Protect persisted secrets at runtime via Keystore (Ch 20).
F5	Enable R8 (shrink/optimize/obfuscate) and retain the mapping file (Ch 83).
F6	Move security-critical logic/secrets server-side.

Non-functional requirements

Attribute	Policy
Security	No extractable high-value secrets; restricted keys; server-side trust.
Maintainability	Secret pipeline reproducible; rotation possible.
Reliability	R8 doesn't break the app (keep rules); mapping retained.
Performance	R8 shrinking reduces size/improves perf.
Honesty	Obfuscation treated as cost-raiser, not protection.

Scope

- **In scope:** secret classification (what can live on client), build-time injection (Gradle, env/secrets, not VCS), restricted third-party keys, runtime protection (Keystore), R8/ProGuard (shrinking/obfuscation/keep rules/mapping), native/string obfuscation caveats, defense-in-depth.
- **Out of scope:** Keystore mechanics (Ch 20), integrity attestation (Ch 26), pinning (Ch 23), crash symbolication details (Ch 83), CI/CD secret stores (general DevOps).

Assumptions & constraints

- Gradle build; R8 (default shrinker). Play App Signing. Secrets come from CI/env/ a secrets manager, **never** committed. `minSdk 24`. **Axiom: an APK cannot keep a secret.**

27.3 System Design Fundamentals

- **An APK cannot keep a secret.** Decompilation + runtime hooking means any value present in the app is extractable. The only robust answer for a *true* secret is: **don't put it in the app** — keep it server-side and expose functionality via authenticated per-user requests (Ch 19).
- **Secret classification.**
  - *True secrets* (grant privileged backend access): **server-side only**.
  - *Restricted keys* (third-party SDKs that require a client key): use provider restrictions (app package + signing cert + API/scope limits) so an extracted key is **useless outside your app** and **revocable/rotatable**.
  - *Non-secret config* (base URLs, feature flags): fine in the app / remote config.
- **Build-time injection, not VCS.** Inject config via Gradle from **environment/CI secrets / a secrets manager / an untracked properties file** — never commit keys. `BuildConfig` fields hold the *injected* value but are still extractable, so only put **restricted/non-secret** values there.
- **Runtime protection (Keystore, Ch 20).** Secrets that must *persist* on device (tokens) are encrypted via Keystore — not stored as plaintext constants.
- **R8 (the modern shrinker/optimizer/obfuscator).** Replaces ProGuard; it **shrinks** (removes unused code/resources), **optimizes**, and **obfuscates** (renames classes/methods to short names). Benefits: smaller/faster app, raised reverse-engineering cost. **It is not encryption** and does not hide strings/logic from a determined attacker. **Keep rules** (`-keep`) preserve classes used via reflection/serialization; **mapping file** must be retained to de-obfuscate crash reports (Ch 83).
- **Native/string obfuscation caveats.** Moving secrets to native (NDK/C++) or encrypting strings raises cost but is still reversible (the code must decrypt at runtime, where it can be hooked). Treat as **cost-raisers**, not protection.
- **Defense-in-depth.** Obfuscation + integrity (Ch 26) + pinning (Ch 23) + Keystore (Ch 20) + server-side enforcement together raise the attacker's cost; none alone is sufficient.

Core insight: **classify secrets and keep true ones server-side; for unavoidable client keys use restricted/revocable ones injected at build time; protect persisted secrets with Keystore; use R8 to shrink and raise cost — never as a secret-keeping mechanism.** Design so the client doesn't hold anything whose leak is catastrophic.

27.4 High-Level Architecture


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Components & ownership

- **Classification** routes each value: server-side / restricted key / non-secret.

- **Build-time injection** supplies restricted/non-secret values from secure sources (never committed).
- **Keystore** (Ch 20) protects persisted secrets at runtime.
- **R8** shrinks/obfuscates and emits a **mapping file** (retained for Ch 83).
- **Defense-in-depth** raises cost around what remains.

Direction: value → classify → (server / restricted-injected / config) → R8 → mapping retained.

27.5 Detailed Component Design

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Element	Responsibility	Note
secret classification	decide placement	server / restricted / non-secret
Gradle injection	build-time config from secrets	not VCS; per-flavor
restricted key config	provider-side restrictions	package+sig+API; revocable
Keystore (Ch 20)	runtime secret protection	tokens, DEKs
R8 rules + mapping	shrink/obfuscate; symbolicate	`-keep` ; retain mapping

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27.6 End-to-End Data Flow

Build-time secret injection + R8

 Build-time secret injection + R8

Runtime: no app secret for backend access

 Runtime: no app secret for backend access

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27.7 Complete Kotlin / Gradle Implementation

27.7.1 Build-time injection (Gradle KTS) — secrets NOT in VCS

```
// build.gradle.kts — read from env/CI or an UNTRACKED local.properties (gitignored).
val mapsKey: String = providers.environmentVariable("MAPS_API_KEY").orNull
    ?: localProps.getProperty("MAPS_API_KEY", "")   // local.properties is .gitignored

android {
    defaultConfig {
        // Restricted key only (package+signature+API limited on the provider side).
        buildConfigField("String", "MAPS_API_KEY", "\"$mapsKey\"")
        // Non-secret config is fine here too:
        buildConfigField("String", "API_BASE_URL", "\"https://api.streamcart.com\"")
    }
    buildFeatures { buildConfig = true }
}
// .gitignore: local.properties, *.keystore, secrets/*   <-- never commit secrets
```

27.7.2 R8 enabled + keep rules + mapping retention

```
android {
    buildTypes {
        release {
            isMinifyEnabled = true            // R8 shrink + obfuscate
            isShrinkResources = true          // remove unused resources
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
                "proguard-rules.pro",
            )
            // mapping.txt is produced in build/outputs/mapping/release/ -> upload to
Crashlytics/Play (Ch 83)
        }
    }
}
```

```
# proguard-rules.pro — keep what reflection/serialization needs.
# kotlinx.serialization
-keepattributes *Annotation*, InnerClasses
-keep,includedescriptorclasses class com.streamcart.**$$serializer { *; }
-keepclassmembers class com.streamcart.** { *** Companion; }
# Retrofit/Gson DTOs accessed reflectively
-keep class com.streamcart.data.**.dto.** { *; }
# Keep line numbers for readable crash stacks (still obfuscate names)
-keepattributes SourceFile,LineNumberTable
-renamesourcefileattribute SourceFile
```

27.7.3 Runtime secret protection (ties to Ch 20)

```
// Persisted secrets (tokens) are NEVER constants; they live encrypted via Keystore (Ch 20).
class ThirdPartySecretProvider @Inject constructor(private val secureStore: SecureStore) {
    // A secret the SDK needs at runtime, fetched from the backend over an authed channel and
    // cached encrypted — not baked into the APK.
    suspend fun sdkToken(): String? = secureStore.get("sdk_token")
}
```

27.7.4 What NOT to do (anti-patterns)

```
// ANTI-PATTERN: hardcoded true secret — extractable from the APK in minutes.
// const val BACKEND_SIGNING_SECRET = "sk_live_9f8a..."   // NEVER

// ANTI-PATTERN: "encrypting" a hardcoded string — the key+logic are also in the APK.
// val key = decrypt("U2FsdGVk...", HARDCODED_KEY)         // reversible at runtime
```

27.7.5 Module / build structure

```
build.gradle.kts           build-time injection (env/untracked props)
proguard-rules.pro         R8 keep rules
local.properties           secrets (GITIGNORED)
secrets/ , *.keystore      GITIGNORED
build/outputs/mapping/...  mapping.txt (upload for symbolication, Ch 83)
# Backend holds all true secrets.
```

27.8 Concurrency and Race Conditions

Secrets/obfuscation are largely build-time, but runtime secret handling has the usual concerns (shared with Ch 20):

Scenario	Risk	Handling
Secret in memory	lingering plaintext	minimize lifetime; zero byte arrays (Ch 20)
Concurrent fetch of a runtime secret	duplicate fetch	single-flight (Ch 11)
R8 breaks reflection at runtime	crash/missing class	correct `-keep` rules; test release build
Mapping file mismatch	unreadable crashes	retain the exact mapping per build (Ch 83)

- Most "races" here are **build correctness** (R8 removing something reflection needs) — test the **release** build, not just debug, and keep rules precise.

27.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Hardcoded secret extracted	security scan / breach	rotate immediately; move server-side; never re-embed	contained if revocable
Restricted key abused outside app	provider logs	rely on package+signature+API restrictions; rotate	limited blast radius
R8 removed a reflected class	release crash	add `-keep` rule; test release	fixed
Mapping file lost	unreadable crashes	retain/upload mapping per release (Ch 83)	symbolicated
Obfuscation assumed to hide a secret	breach	redesign: secret server-side; obfuscation ≠ protection	correct model
Key committed to VCS	git history scan	rotate the key; purge history; add to `.gitignore` + secret scanning	rotated
Over-aggressive optimization bug	runtime error	tune R8 rules; report	stable



> **If a client secret leaks, the fix is rotation + moving it server-side — not "obfuscate harder."** Any client-held secret must be treated as *eventually public*; design for revocation.

27.10 Scaling the Implementation

- **Level 1.** No hardcoded secrets; build-time injection of restricted keys; `.gitignore` + secret scanning.
- **Level 2.** R8 enabled with correct keep rules; mapping retained/uploaded (Ch 83); Keystore for persisted secrets (Ch 20).
- **Level 3 — production.** True secrets server-side (per-user auth, Ch 19); restricted third-party keys with rotation; defense-in-depth (integrity Ch 26, pinning Ch 23).

- **Level 4 — large.** Secrets manager + CI pipeline; automated rotation; per-flavor config; string/native obfuscation for high-value logic (as cost-raisers); commercial obfuscators/RASP where justified.
- **Level 5 — enterprise.** Formal secret governance, SBOM/supply-chain security, key ceremonies, RASP + anti-tamper platforms, continuous reverse-engineering monitoring.

Avoid overengineering: most apps need "no hardcoded secrets + R8 + restricted keys + Keystore." Commercial obfuscators/RASP and native string hiding are for high-value targets (fintech/DRM/anti-cheat) — and still only raise cost.

27.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
True secret placement	Server-side (no app secret)	In-app (encrypted/obfuscated)	**Always** server-side; in-app is extractable
Third-party keys	Restricted + revocable	unrestricted	Restricted always (limit blast radius)
Config injection	Build-time from CI/secrets	committed to VCS	CI/secrets; never commit
Obfuscation	R8 (free, default)	commercial (DexGuard) / RASP	R8 for most; commercial for high-value targets
Hiding logic	Server-side	native/string obfuscation	Server-side for security-critical; native only raises cost
Persisted secret	Keystore-encrypted (Ch 20)	plaintext/BuildConfig	Keystore always

Obfuscation vs server-side (the key trade-off): obfuscation (R8/commercial/ native) raises the *cost and time* of reverse engineering — valuable against casual attackers and for IP protection — but it **never** makes a secret safe, because the code must run (and thus reveal) on the attacker's device. The only way to keep a secret secret is to **not ship it**: put it and the logic that uses it **server- side**. Use obfuscation to slow attackers and shrink the app; use server-side placement to actually protect secrets.

R8 vs commercial obfuscators: R8 is free, integrated, and good for shrinking + name obfuscation; commercial tools (DexGuard) add string encryption, control-flow obfuscation, and RASP (runtime anti-tamper) at cost — justified for high-value apps but still cost-raisers, not guarantees.

27.12 Security Considerations

- **An APK cannot keep a secret** — the foundational rule. True secrets live server-side; the client holds only revocable, scoped, low-value tokens (Ch 19).
- **No hardcoded secrets** anywhere (code/resources/ `BuildConfig` /VCS); use **secret scanning** (gitleaks, GitHub secret scanning) in CI.
- **Restrict and rotate** any unavoidable third-party client key (package + signature + API restrictions).

- **Obfuscation is not encryption** — never rely on R8/string hiding to protect a secret or security-critical check; it only raises cost.
- **Protect persisted secrets with Keystore** (Ch 20), not constants.
- **Defense-in-depth:** combine with integrity (Ch 26), pinning (Ch 23), server-side enforcement; move critical checks server-side rather than hiding them.
- **Play App Signing** keeps the upload/signing key off your machine; protect the signing key.
- **Mapping file is sensitive-ish** (eases reverse engineering) — store it securely, but you must retain it for crash symbolication (Ch 83).
- Client vs backend: the client is public code; the **backend is the only place a secret or security-critical decision is safe**.

27.13 Performance and Resource Optimization

- **R8 improves performance**: shrinking removes unused code/resources (smaller APK, faster load), optimization inlines/simplifies. Enable it in release regardless of the security angle.
- `isShrinkResources` trims unused resources (big size wins).
- **Baseline profiles** (Ch 92) compose with R8 for startup gains.
- **Keep rules cost size** — over-broad `-keep` prevents shrinking; keep them minimal/precise.
- **Native/commercial obfuscation** can add runtime overhead — measure.
- **Profiling:** APK size (before/after R8), method count, startup time (Ch 88), release-build correctness.

27.14 Testing Strategy

The critical tests are **build correctness** (R8 doesn't break the app) and **no-secret** verification.

```
// 1) Release-build smoke/instrumentation tests — catch R8 removing reflected classes.
//    Run the FULL test suite on the minified release build, not just debug.

// 2) Static "no hardcoded secrets" check (CI): fail the build on matches.
//    gitleaks / detekt custom rule / a Gradle task scanning for key patterns.

// 3) Verify BuildConfig holds only restricted/non-secret values.
class BuildConfigTest {
    @Test fun `no true secret in BuildConfig`() {
        assertFalse(BuildConfig.MAPS_API_KEY.startsWith("sk_live"))   // example guard
        // API_BASE_URL is fine (non-secret)
    }
}

// 4) Verify mapping file produced for release (CI check) -> uploaded for symbolication (Ch 83).
// 5) DTO round-trip tests on the minified build (ensures -keep rules correct).
```

What to test: run instrumentation on the **minified release** build; CI secret scanning; mapping-file presence; serialization/reflection still works post-R8.

27.15 Production Observability

- **Metrics:** APK size (R8 effect), crash symbolication rate (mapping uploaded?), restricted-key abuse (provider dashboards), secret-scan findings in CI.
- **Logs:** never log secrets/keys (Ch 28); CI logs for secret-scan results.
- **Debugging:** unreadable crash stacks → mapping file not uploaded/mismatched (Ch 83); release-only crashes → R8 removed a reflected class (missing `-keep`); third-party key abused → restrictions missing/rotate; secret found in a breach → it was in the APK (rotate + move server-side).
- **Alerting:** CI secret-scan hit → block the merge; provider key-abuse alert → rotate.
- **Privacy:** protect the mapping file; never expose it publicly (eases RE).

27.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Can you keep a secret in an APK?* No — it's decompilable and hookable; anything shipped is extractable.
2. *Where do true secrets go?* Server-side; the client gets revocable, scoped tokens (Ch 19).
3. *What is R8?* The default shrinker/optimizer/obfuscator (replaces ProGuard) — shrinks, optimizes, renames; **not** encryption.
4. *Does obfuscation protect secrets?* No — it raises reverse-engineering cost; secrets/logic still run (and reveal) on-device.
5. *Secret classification?* True secret (server-side), restricted key (injected + limited), non-secret config (BuildConfig/remote).

Keys/config

6. *Hardcoded API key — problem?* Trivially extracted; rotate + move server-side.
7. *Unavoidable third-party client key?* Restrict (package+signature+API) + revocable + rotate.
8. *Where to inject build config?* Gradle from env/CI/untracked props — never VCS.
9. *Is `BuildConfig` secret?* No — extractable; only put restricted/non-secret values.

R8

10. *What does `isMinifyEnabled` do?* Enables R8 shrink + obfuscate.
11. *Keep rules?* `-keep` preserves classes used via reflection/serialization.
12. *Mapping file?* De-obfuscates crash stacks — must retain/upload (Ch 83).
13. `isShrinkResources`*?* Removes unused resources (size).
14. *R8 performance?* Smaller/faster app; enable regardless of security.

Runtime protection

15. *Persisted secrets?* Keystore-encrypted (Ch 20), not constants.
16. *String encryption in the APK?* The decrypt key+logic are also in the APK — reversible; cost-raiser only.
17. *Native (NDK) secrets?* Harder but still reversible/hookable.

Model/limits

18. *Obfuscation vs server-side?* Obfuscation slows attackers; server-side actually protects. Use both.
19. *If a client secret leaks?* Rotate + move server-side — not "obfuscate harder."
20. *Defense-in-depth?* + integrity (Ch 26), pinning (Ch 23), Keystore (Ch 20), server enforcement.

Build/VCS

21. *How prevent committed secrets?* `.gitignore` + CI secret scanning (gitleaks).
22. *Play App Signing?* Keeps the signing key off your machine; protect it.

Testing

23. *Why test the release build?* R8 can remove reflected classes → release-only crashes.
24. *CI check for secrets?* Secret scanning fails the build on matches.

Production debugging

25. *Unreadable crash stacks?* Mapping file not uploaded/mismatched.
26. *Release-only crash?* Missing `-keep` rule.
27. *Key abused externally?* Restrictions missing; rotate.

Design-change challenges

1. **Remove a hardcoded backend secret.** Move to server-side auth (Ch 19); rotate; the app no longer needs it.
2. **Third-party SDK requires a client key.** Restricted+revocable key injected at build time.
3. **Protect business logic from cloning.** Server-side critical logic + integrity (Ch 26); obfuscate the rest.
4. **Crashes unreadable after enabling R8.** Retain/upload mapping (Ch 83).
5. **High-value app (fintech/DRM).** R8 + commercial obfuscation/RASP + integrity + server-side — as layered cost-raisers.

Mock walkthrough (abridged)

Clarify (what values, which truly secret) → "APK can't keep a secret" → classify (server-side / restricted / non-secret) → build-time injection (not VCS) → Keystore for persisted (Ch 20) → R8 for shrink + cost (keep rules + mapping) → defense-in- depth (Ch 26/23) → state obfuscation ≠ protection → testing (release build + secret scan).

27.17 Practical Exercises

1. **Beginner.** Move a hardcoded key to build-time injection from an untracked source; verify it's not in VCS. *Eval:* no key in git; injected at build.

2. **Intermediate.** Enable R8 + `isShrinkResources` with correct keep rules for kotlinx.serialization/Retrofit; run the suite on the release build. *Eval:* app works minified; mapping produced.
3. **Advanced.** Replace an in-app "app secret" for backend access with per-user auth (Ch 19) so no app secret exists; restrict a remaining third-party key. *Eval:* no extractable high-value secret; restricted key.
4. **Production debugging.** Fix release-only crashes (missing `-keep`) and unreadable stacks (no mapping upload, Ch 83). *Eval:* release stable; crashes symbolicated.
5. **System design challenge.** Design a secret pipeline (CI secrets → Gradle injection → restricted keys → Keystore runtime) with rotation and secret scanning. *Eval:* no committed secrets; rotatable; scanned.

(Solutions withheld unless requested.)

27.18 Chapter Summary & Reference Sheet

Key decisions: treat the **APK as public** — **keep true secrets server-side** and give the client only revocable, scoped tokens (Ch 19); for unavoidable third-party client keys use **restricted + revocable** keys **injected at build time from secure sources (never VCS)**; **protect persisted secrets with Keystore** (Ch 20), not constants; enable **R8** (shrink/optimize/obfuscate) for size/perf and to **raise reverse-engineering cost**, with precise **keep rules** and a **retained mapping file** (Ch 83); use **defense-in-depth** (integrity Ch 26, pinning Ch 23) and **move critical logic server-side**; **obfuscation is a cost-raiser, not encryption**; if a client secret leaks, **rotate + move server-side**.

Concepts: APK is decompilable, secret classification, server-side secrets, restricted/revocable keys, build-time injection, R8 shrink/obfuscate/keep-rules/ mapping, obfuscation ≠ encryption, defense-in-depth, rotation.

Common mistakes: hardcoding secrets (code/resources/BuildConfig/VCS); treating obfuscation as protection; "encrypting" a hardcoded string; unrestricted third-party keys; committing secrets; disabling R8 (size/perf loss) or enabling it without keep rules/mapping; hiding security-critical logic in the client instead of the server.

Checklist:

- [ ] no hardcoded true secrets
- [ ] server-side true secrets
- [ ] restricted+ revocable third-party keys
- [ ] build-time injection (not VCS) + secret scanning
- [ ] Keystore for persisted secrets
- [ ] R8 enabled + keep rules
- [ ] mapping retained/uploaded
- [ ] defense-in-depth
- [ ] rotate on leak
- [ ] test release build.

Key APIs/tools: Gradle `buildConfigField`/env providers, R8 (`isMinifyEnabled`, `isShrinkResources`, `proguardFiles`, `-keep`), `mapping.txt`, Play App Signing, Keystore (Ch 20), gitleaks/secret scanning, Firebase Crashlytics mapping upload (Ch 83), DexGuard/RASP (commercial).

References:

- Shrink/obfuscate with R8: https://developer.android.com/build/shrink-code

- Keep rules / ProGuard config: https://developer.android.com/build/shrink-code#keep-code
- Play App Signing: https://developer.android.com/studio/publish/app-signing#app-signing-google-play
- Securing API keys / restrictions: https://developer.android.com/privacy-and-security/security-tips#UserData
- (composes with) Ch 20 Keystore, Ch 23 Pinning, Ch 26 Integrity, Ch 83 Crash Reporting

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/flow/failure diagrams; Gradle injection + R8 config + keep rules + runtime provider + anti-patterns; concurrency (build correctness); failure (leak→rotate, R8 removal); 5 levels; obfuscation-vs-server-side and R8-vs-commercial trade-offs; "APK can't keep a secret" model; perf (R8 size/speed); release-build + secret-scan tests; observability (symbolication); 27 Q&A + 5 challenges. ✅
- **Expand later:** crash symbolication (Ch 83), baseline profiles (Ch 92), integrity (Ch 26), RASP/commercial obfuscation.
- **Interview coverage:** exceeds ≥25 bar.